

TML2026: MEMBERSHIP INFERENCE ATTACK (ASSIGNMENT 1)

Madhvan Bajaj
Msc Data Science and AI
Saarland University
7072049

Anish Chandrasekaran
Msc Data Science and AI
Saarland University
7072812

1 GITHUB LINK

https://github.com/madhvanb/TML_Assignment.git

2 INTRODUCTION

The task was to decide whether for each image it was part of the training set as a part of a ResNet-18 classifier. For each sample that was provided as a part of the file `priv.pt`, we output a score in the range of 0 and 1. Here the higher the score it makes it more likely that it is a member. The evaluation metric is True Positive Rate at 5% False Positive Rate, so basically our goal is to not just separate members from the non-members but to rank true members high while also keeping the false alarm generated low.

3 MAIN BODY

We began with a perturbation-based baseline that measured how stable the model's true-label confidence remained under small image transformations. This was fast and gave an initial score of about 0.056, but it was still an indirect proxy for memorisation. Two samples can behave similarly under augmentation even if only one of them was actually used in training. We therefore moved to a shadow-model attack based on LiRA (1), which estimates for each sample how likely its target-model behaviour is under an in-training versus out-of-training distribution.

Our final method trains 32 ResNet-18 shadow models for 25 epochs each on random halves of the merged `pub.pt` and `priv.pt` pool. We use SGD with momentum, Nesterov acceleration, weight decay, cosine annealing, and standard image augmentation through random crops and horizontal flips. For each sample, we collect the logit of the true-class probability from shadow models that did and did not train on it, fit Gaussian distributions to those two groups, and compute a log-likelihood ratio for the target model.

$$s(x) = \log p(z(x) \mid \text{in}) - \log p(z(x) \mid \text{out}).$$

We then apply an RMIA ranking step (2) that converts each score into a classwise percentile against public non-members of the same class. This classwise calibration matters because the metric operates at 5% FPR, where miscalibration across classes can hurt badly.

Our biggest decision was to probably combine the private sets for training shadows. Previously we only trained shadows using the public set but this came at the cost of the private set not having real observations during the training phase. Due to this their scores were based on the class-level fallbacks. We then decided to change this by merging the two sets and training about 32 shadows while also keep the RMIA calibration step. This change gave us a score of 0.062130 and a rank of 4. This is a definite improvement from our initial results that we had observed which was around 0.056 and a rank of 20. The shadow training method was really important for us and merging made a considerable difference.

4 CONCLUSION

Membership inference attack can basically reveal if a certain record was used while training a model. This is an important consideration because if the record was used in training this could be some important information. When we deal with real world data which could be something like financial information we truly need to be careful. From our observation a shadow model attack can be used to figure out if a record was used or not. This raises an important consideration of privacy that we should consider during the model building phase itself rather than an afterthought when we are done creating a model.

REFERENCES

- [1] Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramèr. Membership Inference Attacks From First Principles. In *2022 IEEE Symposium on Security and Privacy (SP)*, pp. 1897–1914, 2022. <https://ieeexplore.ieee.org/document/9833649/>
- [2] Sajjad Zarifzadeh, Philippe Liu, and Reza Shokri. Low-Cost High-Power Membership Inference Attacks. arXiv:2312.03262, 2023. <https://arxiv.org/abs/2312.03262>